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A New Model for Scheduling Operations in Modern Agricultural Processes

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Abstract. In recent years, the increase in population and the decrease in agricultural lands and water shortages have caused many problems for agriculture and farmers. That is why scheduling is so important for farmers. Therefore, the implementation of an optimal schedule will lead to better use of agricultural land, reduce water consumption in agriculture, increase efficiency and quality of agricultural products. In this research, a scheduling problem for harvesting agricultural products has been investigated. In this problem, there are n number of agricultural lands that in each land m agricultural operations are performed by a number of machines that have different characteristics. This problem is modeled as a scheduling problem in a flexible workshop flow environment that aims to minimize the maximum

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completion time of agricultural land. The problem is solved by programming an integer linear number using Gams software. The results show that the proposed mathematical model is only capable of solving small and medium-sized problems, and due to the Hard-NP nature of the problem, large-scale software is not able to achieve the optimal solution.

Keywords: Scheduling; Agricultural operations; Flexible workshop flow; Maximum completion time.

1. Introduction

Crops are very diverse, used in different ways and in agricultural operations are very different from each other. However, different crops have a lot in common. In general, crops can be grouped in different ways, including based on botanical and evolutionary characteristics, purpose of production and consumption, plant longevity, environmental needs and favorable growing conditions, crop operations, etc. Crops may be divided into two or more different groups due to their different characteristics and multi-purpose production [1].

In today's world, natural resources are being destroyed more and more, so that rapid population growth in recent decades, changing lifestyle, rural migration to cities, low level of technology, difficult ecological conditions, limited suitable agricultural lands, lack of resources Water, lack of proper use of basic resources and insufficient investment have intensified the process of resource degradation [2].

Accordingly, sustainable development in order to manage and protect basic resources and the introduction and application of technical advances and appropriate organizational structure in this area recommends measures [3]. Meanwhile, the discussion on agriculture and sustainable development in this economic sector due to its importance in various economic, social, political sectors and the sensitive role of basic and natural resources in this sector is of fundamental importance and priority [4].

Undoubtedly, the issue of food and food security is one of the most fundamental and important challenges of the present and the future [3]. On the one hand, the increase in world population and the use of the maximum area of arable land, has led the human movement from trying to increase the area under cultivation to increase yield per unit area. Therefore, agricultural scientists have for years chosen the two issues of "farming" and "breeding" as the main and general strategies to ensure the future of human food. In this regard, plant breeding specialists have turned to producing improved and high-yielding cultivars, geneticists and agricultural biotechnology scientists have turned to plant gene manipulations, and agronomists have turned to improving existing cropping systems and inventing new farm management systems. Among them is the issue of water scarcity and understandable sensitivities that have been created in recent years on the issue of environmental protection and conservation of energy resources, and considering that traditional agricultural systems with mismanagement of water and excessive use of fertilizers, herbicides and pests Chemical pesticides have been one of the most important sources of environmental pollution [4].

Therefore, agronomists in recent years have been looking for new methods in farm management that in addition to optimizing input consumption, also increase yield and ultimately increase the economic efficiency of production. It was to this end that the issue of precision agriculture has been raised since the beginning of the last decade. Obviously, at the level of a farm, however small, we encounter variable variables [5].

Scheduling is a decision-making process used regularly in most industrial and service industries. In fact, scheduling deals with the allocation of resources to tasks over time and its goal is to optimize one or several objectives [6]. There are three parts of sowing, growing, and harvesting in agricultural operations, each involving several different tasks each being performed in agricultural lands by various machinery and human resources. Scheduling in agricultural operations contributes to the proper management of all agricultural lands and machinery and human resources. In addition, scheduling leads to the completion of an optimal series of agricultural operations in agricultural lands that exist in several fields with various and scattered areas using bounded machinery and human resources in the shortest time possible [1].

Considering the issues related to sustainable agricultural development, it is important to note that achieving sustainable development in this economic sector can not be achieved except by examining various aspects and proper management of various dimensions.

Scheduling even increases productivity and product quality and decreases water use and time to work on agricultural lands. Moreover, it increases profits and decreases costs in agricultural operations. Elderen studied agricultural operation scheduling models and techniques. This scholar focused on three solution techniques, aiming at determining the best option among the three to carry out better agricultural scheduling for different agricultural operations [7]. Ferrer et al. assessed an optimization approach for optimally scheduling wine grape harvesting operations taking into account both operational costs, grape quality, and routing costs [8]. In another research, Guan et al. proposed a resource assignment and scheduling based on a two-phase metaheuristic for a long-term cropping schedule. They discussed two components of resource allocation optimization and optimal scheduling and sequencing for a hybrid parallel systems model [9]. Bochtis et al. evaluated a flow-shop problem formulation of biomass handling operations scheduling. They applied specific programming models and formulating these models as best as possible [10]. Orfanou et al. presented a planning approach for scheduling sequential tasks involved in biomass harvesting and handling operations performed by machinery teams. Their approach determined in which fields each machine had to operate, in what sequence, in which period of time [11].

Thuankaewsing et al. assessed sugar cane harvest scheduling to guarantee the profit of cane growers. These scholars mainly aimed to determine the most suitable sugar cane harvest schedule for a cluster of sugar cane fields in order to harvest the highest yield and high quality of sugar cane during the harvest season [12,16]. Edwards et al. (2015) evaluated optimized schedules for consecutive agricultural operations and assessed simultaneous scheduling of multiple machines by performing multiple operations in different areas [13]. Aguayo et al. addressed a corn-stover harvest scheduling problem (CSHSP) that arises when a cellulosic ethanol plant contracts with farmers to harvest corn stover after the grain harvest has been completed [14]. D'Urso et al. considered the problem of refill scheduling for a team of vehicles or robots that must contend for access to a single physical location for refilling. These scholars studied the time spent to refill fuel tanks and tanks and time lost in the queue [15].

As mentioned, the problem of scheduling in agricultural harvesting operations is modeled as a scheduling problem in a flexible workshop flow environment in which agricultural fields are considered as jobs, agricultural operations are recognized as work processes, and machinery for as the machines of each operation for each operation. The objective function is to minimize the maximum operation completion time. The remainder of this paper is structured as follows: the second section presents the problem and mathematical model while the third section provides the solution of the mathematical model. Ultimately, the fourth section concludes and makes suggestions for future studies.

2. Problem Statement

The present study aims to propose a model in order to determine the sequence of operations and scheduling of harvest operations in several farms with different sizes in geographically dispersed conditions. The problem is modeled as a scheduling problem in a flexible workshop flow environment by a number of available machinery and human resources. In this work environment, tasks are lands, in which a number of operations are carried out by different types of machinery. Technical assessments of the machinery are performed by farmers in order to determine which operations are carried out by which machinery. In addition, there are different operations such as harvesting, gathering, packing, and loading in each farm, performed by a special machine for the same operation. It is notable that the machinery and human resources cannot leave the farm or move to another farm until all operations are completed on each farm. In addition, the operations existing in each farm are different in terms of tasks performed by the machinery. Moreover, a central warehouse is considered for the machinery in order to move toward the farm from the warehouse at the beginning of the work and then go back to the warehouse at the end of the working shift. Furthermore, a rest time is considered for employees, during which no operation is carried out.

2.1. Mathematical Model

This section introduces the indexes, parameters, problem decision variables, and mathematical model of the linear integer of the research.

Indexes

j= Index of agricultural lands

i= Index of operations

k= Index of machinery

Parameters

n= Number of agricultural lands

m= Number of operations; m=1,2,3,4

s_{ik}= The series of k machinery for i operation

v_{ik}= Speed of operations for i operation by k machinery

D_{lj}= Distance between l and j

NP_{jki}= Duration of primary processing of j agricultural land on k machinery in i operation

U= The number of human resources

A= Processing time reduction coefficient in i operation due to human resource allocation

R_j= Maximum human resources that can be allocated to j tasks in the final operations

M= A very big number

Decision Variables

C_{max}= Maximum completion time of agricultural lands

ST_{ji}= Start time of j agricultural land in i operation

C_{ji}= Completion time of j agricultural land in i operation

P_{jki}= Final processing time of j agricultural land in i operation by k machinery

r_j= The number of human resources allocated to j agricultural land in the final operation

S_{jki}= The processing time of j agricultural land in i operation by k machinery

B_{jli}= The number of human resources allocated to j and l agricultural lands that overlap in i operation

ρ_{jli}= 1, if the completion time of j agricultural land is equal to or larger than the start time of i operation in l land; otherwise, 0.

X_{jki}= 1, if i operation is performed by k machinery; otherwise, 0.

Y_{lji}= 1, if i operation of the l land is performed before operations of j land; otherwise, 0.

Ya_{ji}= 1, if the completion time of j agricultural land in i operation is equal to or smaller than the rest time; otherwise, 0.

Yb_{ji}= 1, if the start time of i operation in j agricultural time is equal to or smaller than the rest time; otherwise, 0.

$$Min\ Z = C_{max} \tag{1}$$

$$\sum_{k=1}^{s_{ik}} X_{jki} = 1 \quad \forall j, i, j \geq 2 \tag{2}$$

$$C_{jl} \geq \sum_{k=1}^{s_{lk}} S_{jkl} + \sum_{k=1}^{s_{lk}} \frac{D_{0j}}{v_{lk}} X_{jkl} \quad \forall j; j \geq 2 \tag{3}$$

$$C_{ji} \geq C_{j,i-1} + \sum_{k=1}^{s_{ik}} S_{jki} \quad \forall j, i; j \geq 2, i \geq 2 \tag{4}$$

$$C_{ji} + M(3 - X_{jki} - X_{lki} - Y_{lji}) \geq C_{li} + S_{jki} + \frac{D_{lj}}{v_{ik}} X_{jki} \quad \forall j, l, k, i; l \neq j; j \geq 2 \tag{5}$$

$$C_{li} + M(2 - X_{jki} - X_{lki} - Y_{lji}) \geq C_{ji} + S_{lki} + \frac{D_{jl}}{v_{ik}} X_{lki} \quad \forall j, l, k, i; l \neq j; j \geq 2 \tag{6}$$

$$-C_{ji} + 14 \leq M\ ya_{ji} \quad \forall j, i; j \geq 2 \tag{7}$$

$$C_{ji} - 12 \leq M\ (1 - ya_{ji}) \quad \forall j, i; j \geq 2 \tag{8}$$

$$C_{ji} \leq 12 + M\ (1 - yc_{ji}) \quad \forall j, i; j \geq 2 \tag{9}$$

$$C_{max} \geq max \left\{ C_{jm} + \sum_{k=1}^{s_{mk}} \frac{D_{j0}}{v_{mk}} \right\} \quad \forall m, j; j \geq 2 \tag{10}$$

$$C_{ji} - \sum_{k=1}^{s_{ik}} S_{jki} = ST_{ji} \quad \forall j, i; j \geq 2 \tag{11}$$

$$-ST_{ji} + 14 \leq\ yb_{ji} \quad \forall j, i; j \geq 2 \tag{12}$$

$$ST_{ji} - 12 \geq M(1 - yb_{ji}) \quad \forall j, i; j \geq 2 \tag{13}$$

$$-ST_{ji} \leq -14 + M\ yc_{ji} \quad \forall j, i; j \geq 2 \tag{14}$$

$$P_{jki} = NP_{jki} - Ar_{j4} \quad \forall j, k, i; j \geq 2 \tag{15}$$

$$r_{j4} \leq R_{j4} \quad \forall j; j \geq 2 \tag{16}$$

$$M\ \rho_{jl4} \geq C_{j4} - C_{l4} - \sum_{k=1}^{s_{4k}} S_{lk4} \quad \forall j, l, i; j \neq l; j \geq 2 \tag{17}$$

$$r_{j4} + B_{jl4} \leq U \quad \forall j, l; j \neq l; j \geq 2 \tag{18}$$

$$S_{jki} + M(1 - X_{jki}) \geq P_{jki} \quad \forall j, k, i; j \geq 2 \tag{19}$$

$$S_{jki} \leq M\ X_{jki} \quad \forall j, k, i; j \geq 2 \tag{20}$$

$$S_{jki} \leq P_{jki} \quad \forall j, k, i; j \geq 2 \tag{21}$$

$$B_{jl4} + M(2 - \rho_{jl4} - \rho_{lj4}) \geq r_{l4} \quad \forall j, l; l \neq j; j \geq 2 \tag{22}$$

$$B_{jl4} \leq M(\rho_{jl4} + \rho_{lj4} - 1) \quad \forall j, l; l \neq j; j \geq 2 \tag{23}$$

$$B_{jl4} \leq r_l \quad \forall j, l; l \neq j; j \geq 2 \tag{24}$$

$$C_{ji}, ST_{ji}, S_{jki}, B_{jli}, P_{jki}, r_{ji} \geq 0 \tag{25}$$

$$X_{jki}, Y_{lji}, \rho_{jli}, ya_{ji}, yb_{ji}, yc_{ji} \in \{0,1\} \tag{26}$$

Constraints (1) introduce the minimization of maximum agricultural land completion time as an objective function. In addition, Constraints (2) express that each agricultural land must be processed by only one machinery in each operation. Constraints (3) mention that the completion time of the first agricultural land is equal to or larger than the processing time of that agricultural land plus the distance traveled from the machinery warehouse to that agricultural land. Constraints (4) show that the completion time of each agricultural land is equal to the completion time of that agricultural land in the previous operation plus the processing time of that agricultural land in that operation. Constraints (5 and 6) calculate the completion time based on the position of each agricultural land in each operation and determine the sequence of agricultural lands. Constraints (7-9) express that no agricultural land must be completed during the rest time. Constraints (10) demonstrate that the maximum agricultural land completion must be equal to or larger than the maximum agricultural land completion in the last operation plus the distance traveled to the machinery station. Constraints (11) express that the agricultural land start time must be equal to the completion time of the operation in that agricultural time minus the processing time in the agricultural land. Moreover, constraints (12-14) mark that no task must be started during the rest time. Constraints (15) calculate the modified processing time of each agricultural land after the allocation of resources to agricultural land. Moreover, constraints (16) express the maximum resources that can be allocated to each operation in each agricultural land. Furthermore, constraints (17) determine the overlapped agricultural lands. Constraints (18) express that the total resources used for agricultural land that are processed in one operation simultaneously cannot exceed the maximum resources in that operation. In addition, constraints (19-21) express the processing time of land by a machine in that operation, which is expressed to linearize the model. Constraints (22-24) show the number of resources allocated to the agricultural land and the next one in the operation if the agricultural lands overlap, which is expressed for model linearization. Ultimately, constraints (25 and 26) determine the range of decision variables.

3. Computational Results

This section evaluates the efficiency of the mathematical model presented in the previous section. To this end, eight experimental problems with different dimensions in small and medium sizes are generated based on the information in Table 1.

Table 1. Variable range of parameters

Type of Parameter	Processing Time	Number of Tasks (Agricultural Land)	Number of Machineries	Number of Operations	Distances between Lands
Range of Parameter	[2 20]	[3 15]	[1 12]	[4]	[1 15]

Table 2. Problems solved by GAMS software

Column	Number of Lands	Operations (Stage) / Machinery				Completion Time (Minute)	Error	Solution Time (Second)
1	3	1	2	3	4	102	0	0.236
		1	1	1	1			
2	4	1	2	3	4	105.800	0	0.187
		2	1	1	1			
3	5	1	2	3	4	109.167	0	0.281
		1	2	1	1			
4	6	1	2	3	4	124.733	0	0.719
		1	2	2	1			
5	8	1	2	3	4	118.00	0	4.125
		3	2	1	2			
6	10	1	2	3	4	146.100	0	30.329
		2	1	2	3			
7	12	1	2	3	4	171.233	10.35	1000.063
		1	2	3	3			
8	15	1	2	3	4	203.500	42.500	1000.093
		2	3	1	4			

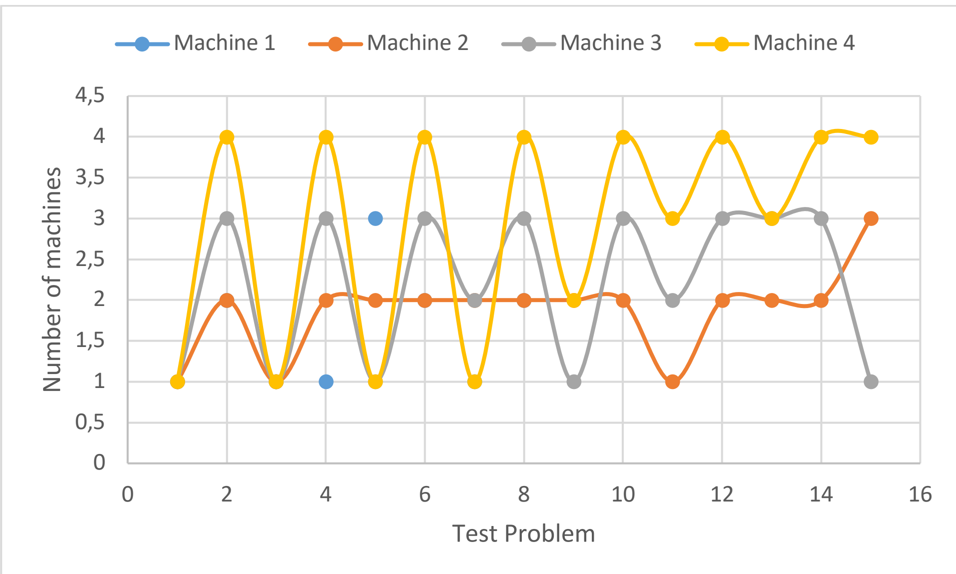

Figure 1. The optimal number of machines in each test problem

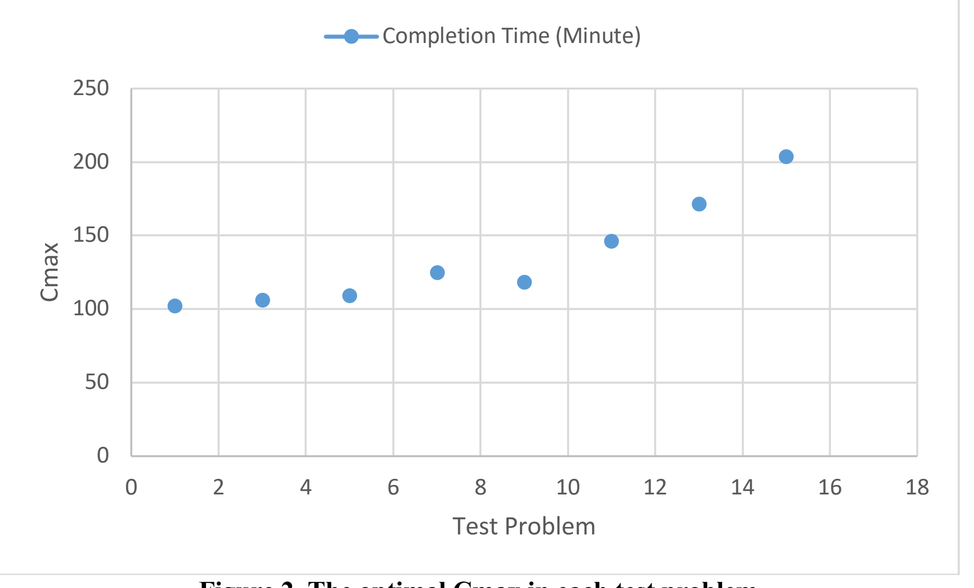

Figure 2. The optimal Cmax in each test problem

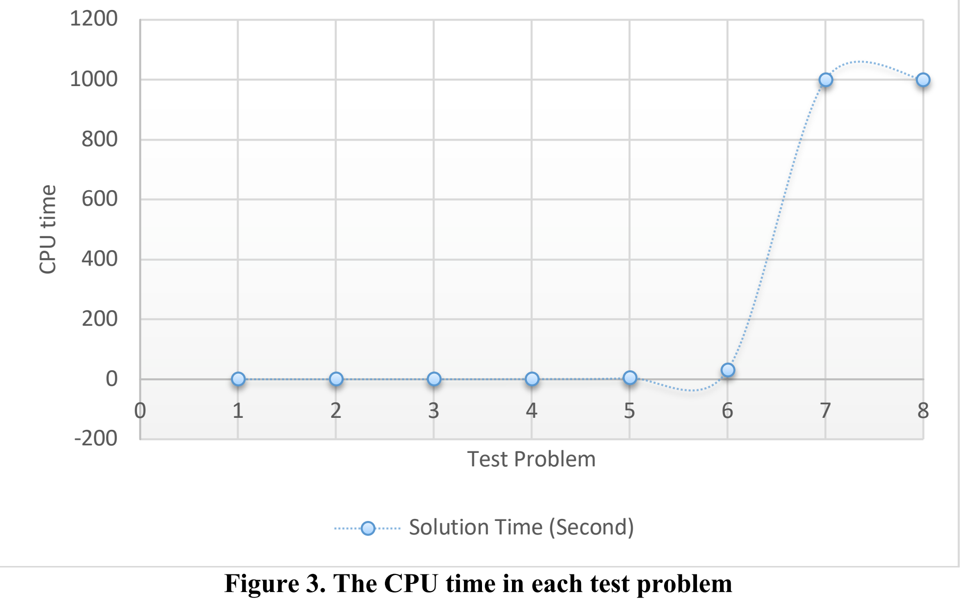

Figure 3. The CPU time in each test problem

As can be seen in Figure 1, the number of machines in problem testing has been quite variable so that its increase and decrease has not been predictable. This is one of the optimization functions that can predict the best amount of variables in addition to the number of predictions. On the other hand, Figure 2 shows that there is an uptrend in Cmax. The reason for this increase is that with the scale of the problem, it is expected that more time will be needed for agricultural activities. The same is confirmed in Figure 3. Due to the increasing dimensions of the problem, the solution time of the mathematical model has also increased sharply

As mentioned in the previous sections, the problem is an NP-Hard one, and the results presented in Table 2 demonstrate that the mathematical model can only solve small-size problems. As observed, as the number of dimensions of the problem increases, the mathematical model is not able to provide the optimal solution. This result shows the high complexity of the research problem and the need for other methods to solve this problem.

4. Discussion and Conclusion

In this study, we evaluated a scheduling problem for harvesting agricultural products in a flexible workshop flow. The presented model was solved in GAMS software. In addition, experimental problems with different sizes (small and medium) were generated and solved based on various parameters. According to the results, the mathematical model presented was only able to solve small-size problems and was unable to reach an optimal solution for medium and large problems. It is recommended that heuristic and metaheuristic methods be applied in future studies. In addition, it is suggested that assumptions such as uncertainty be considered in other fields.

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